

SACHEON AB, Korea, Republic ofWMO: **471610**Lat: **35.089N**Lon: **128.070E**Elev: **26**StdP: **14.68**Time Zone: **9.00 (E09)**Period: **82-19**WBAN: **43240****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.9	16.2	3.1	6.5	27.4	6.7	7.8	25.8	16.7	34.3	14.2	35.6	1.7	60	0.397

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.8	91.6	79.7	89.4	78.4	86.4	76.9	80.7	88.7	79.8	86.9	78.8	84.9	7.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.8	149.7	85.6	77.4	142.5	83.8	76.9	140.3	83.2	44.5	89.2	43.4	87.2	42.3	85.2	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.6	13.4	11.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.4	31.8	35.5	44.2	55.1	63.8	71.0	77.8	79.6	70.5	59.7	48.6	37.7		
	DBStd	16.85	4.86	6.82	5.58	5.95	4.17	3.58	4.44	3.63	5.16	5.71	6.07	5.94		
	HDD50	1662	564	406	189	19	0	0	0	0	0	5	96	383		
	HDD65	4403	1029	825	644	302	72	3	0	0	11	177	493	847		
	CDD50	3994	0	1	10	172	427	631	860	919	614	306	53	1		
	CDD65	1260	0	0	0	4	34	185	396	454	174	13	0	0		
	CDH74	10378	0	0	0	25	318	1189	3335	4217	1163	131	0	0		
	CDH80	3272	0	0	0	1	24	257	1130	1609	247	4	0	0		
	WSAvg	4.3	3.4	4.0	5.0	5.4	5.0	4.5	5.1	4.3	3.7	3.5	3.6	3.6		
	PrecAvg	62.3	1.1	2.0	3.3	5.5	6.0	7.5	12.3	12.0	7.1	2.4	2.0	1.1		
(15) Precipitation	PrecMax	90.0	2.4	5.4	6.5	11.7	11.6	14.4	27.1	29.8	17.6	7.7	7.7	4.0		
	PrecMin	34.3	0.0	0.0	0.6	1.2	2.1	1.9	2.3	3.1	0.6	0.0	0.0	0.0		
	PrecStd	14.2	0.6	1.4	1.6	2.6	2.4	4.1	6.4	7.2	5.4	1.9	1.9	1.1		
	PrecStd	14.2	0.6	1.4	1.6	2.6	2.4	4.1	6.4	7.2	5.4	1.9	1.9	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.3	60.7	66.1	77.0	82.6	87.9	95.4	95.1	88.2	80.5	70.2	61.2		
		MCWB	44.5	51.8	52.7	63.3	66.5	71.6	79.5	81.0	76.5	65.6	61.3	53.1		
	2%	DB	49.5	55.1	62.3	73.5	80.3	85.7	91.5	92.3	85.6	77.3	67.6	57.2		
		MCWB	41.6	47.8	51.4	60.3	65.4	70.7	79.6	79.9	74.6	65.0	57.4	49.3		
	5%	DB	46.3	50.9	59.2	70.3	77.4	82.8	88.2	89.9	82.6	75.0	64.5	53.8		
		MCWB	39.1	43.4	49.2	58.5	64.0	70.0	78.9	78.8	72.4	63.8	54.8	45.9		
	10%	DB	43.0	48.2	56.8	67.9	75.3	80.7	86.1	87.9	80.5	71.8	61.1	50.2		
		MCWB	36.3	42.4	47.6	57.9	62.4	69.9	77.9	78.4	71.4	61.6	53.6	43.2		
	0.4%	WB	46.5	55.8	57.6	67.5	71.5	76.4	81.9	82.8	79.1	71.4	64.9	54.7		
		MCDB	50.7	58.4	61.6	71.5	77.6	81.9	90.1	91.6	85.0	76.4	68.0	60.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	42.3	48.3	53.8	64.3	69.1	74.6	80.6	81.0	77.6	67.9	59.8	50.6		
		MCDB	47.5	50.9	59.4	69.9	74.4	78.9	88.7	89.2	82.9	73.2	64.6	56.1		
	5%	WB	39.7	45.0	51.0	61.5	67.3	73.0	79.7	80.1	75.9	65.4	57.4	47.3		
		MCDB	44.2	49.8	56.5	67.1	73.4	78.6	86.8	87.5	80.8	71.6	62.1	51.8		
	10%	WB	37.5	42.4	48.7	59.2	65.3	71.7	78.7	79.2	73.4	63.4	54.7	44.3		
		MCDB	42.9	46.9	54.4	66.0	71.9	78.3	84.8	85.8	78.3	69.9	60.5	49.1		
	5% DB	MDBR	22.9	21.4	22.4	22.4	20.8	17.3	13.0	14.8	17.9	23.4	23.6	24.3		
		MCDBR	25.2	24.9	26.2	25.7	25.4	23.3	17.5	18.4	19.4	25.8	26.3	27.9		
		MCWBR	18.4	17.3	16.5	14.2	13.1	11.0	7.8	8.2	9.7	14.3	16.9	20.0		
		MCWBR	22.7	20.8	22.2	19.7	19.1	16.0	15.2	15.7	14.6	20.8	21.8	26.6		
(35) Mean Daily Temperature Range	5% WB	MCWBR	17.8	15.6	16.1	13.3	11.8	9.3	7.5	7.8	8.7	13.2	15.7	20.3		
		MCWBR	17.8	15.6	16.1	13.3	11.8	9.3	7.5	7.8	8.7	13.2	15.7	20.3		
	Clear-Sky Solar Irradiance	taub	0.427	0.477	0.557	0.585	0.599	0.651	0.612	0.568	0.507	0.471	0.443	0.421		
		taud	2.010	1.923	1.759	1.748	1.763	1.724	1.867	1.973	2.084	2.059	2.046	2.041		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	233	235	227	230	229	217	224	232	240	235	226	225		
		Edn at Noon	43	53	68	72	71	74	64	56	48	45	41	40		
(44) All-Sky Solar Radiation	RadStd	RadAvg	856	1074	1394	1593	1748	1540	1427	1457	1272	1172	878	773		
		RadStd	64	98	121	125	185	140	207	187	135	107	96	45		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page